

(i) 4 = left auricle & 5 = Right auricle

(ii) Arteries carry blood away from the heart

(iv) Because blood enters the heart twice in a single circulation. ~~One~~ The right side pumps deoxygenated blood and the left side pumps oxygenated blood.

CHEMISTRY

Q1A) Given :-
$$\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 \downarrow + 2\text{NaCl}$$

the ppt)

So the precipitate formed is BaSO_4 which is white in color.

Q2A) Ammonium sulphate is warmed with NaOH solution to give out ammonia gas.

Q11A) (a) The alveoli of lungs are covered with blood capillaries ~~so that~~ because it allows gaseous exchange to happen more freely, directly into the blood stream.

(b) The wall of the trachea is supported by cartilaginous rings as it prevents the trachea from collapsing.

Q12A) (a) (i) As metabolic activity is less in winters, less water is reabsorbed.
(ii) We also sweat much less as compared to other seasons.
(iii) Therefore more urine is produced in winter.

(b) Eyes, skin, lungs etc, Eyes get rid of mucus, skin excretes sweat, lungs give out CO_2 etc.

(c) The malpighian tubules are the main excretory organs of Arthropoda.

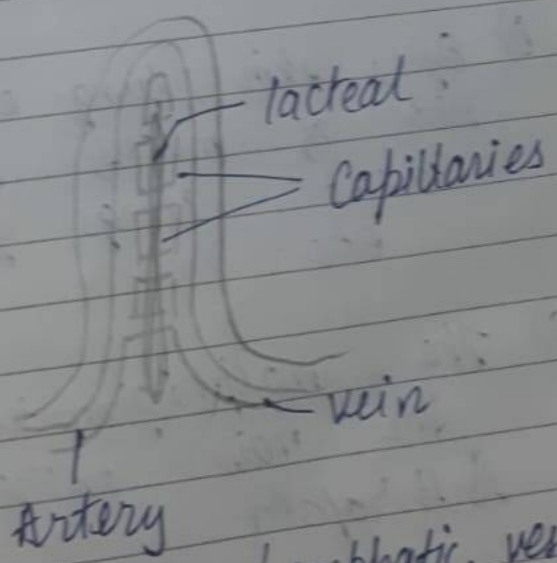
Q4) There are many enzymes that act on proteins. The major ones are pepsin and trypsin.

Q In the absence of O_2 pyruvate will be converted to lactic acid releasing little ATP.

Q Role of Acid in our stomach

- ⇒ Helps soften the food
- ⇒ Provides acidic medium for conversion of pepsinogen and prorenin into active forms of pepsin and renin.
- ⇒ kills microorganisms
- ⇒ stops action of salivary amylase.

Q villi



Lacteal are lymphatic vessels that absorb fats

Q7A) Unlike animals, plants do not have special organs for gaseous exchange, they have small pores in their leaves called stomata and lenticels which help in exchange of gases.

(i) The CO_2 released during respiration is later utilized for photosynthesis but animals don't utilize the CO_2 . ~~Plants can also~~

(ii) Plants can also make their own glucose.

Q8A) Light Reaction

(i) It is dependant on sunlight

(ii) In this process water molecules are split into H_2 and O_2

(iii) Oxygen is released

(iv) Produces ATP

Dark Reaction

(i) It is not dependant on light

(ii) The reduction of CO_2 into carbohydrates

(iii) O_2 is not released

(iv) Produces $\text{C}_6\text{H}_{12}\text{O}_6$

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BIOLOGY

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- Q11) The substance that connects light reaction and dark reaction is ATP.
- Q12) Aerobic respiration in eukaryotic cells occurs in Mitochondria.
- Q13) The lymph is devoid of RBC [red blood cells].
- Q14) Plants get rid of excess water through transpiration, some of these are also removed when dead leaves and ripe fruits fall off the trees.
- Q15) Malpighian body is formed by glomerulus and Bowman's capsule.
- Q16) Options not given, but, Transpiration is affected by, humidity, wind speed, light, temperature and area of the leaf.

Anything except the above factors do not contribute towards rate of transpiration.